F78PA Probability & Statistics A

Tutorial 7 – Solutions

1. Two random variables X and Y have joint probability mass function $p_{X,Y}(x,y)$ given by the following table

	$y=-1$	$y=0$	$y=1$
$x=-1$	0	0.25	0
$x=0$	0.25	0	0.25
$x=1$	0	0.25	0

Determine the following:

(a) Calculate $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.
We can obtain the marginal distribution of X by summing the rows, this gives that $X=-1$ with probability 0.25, $X=0$ with probability 0.5 and $X=1$ with probability 0.25. Therefore the expectation of X is $\mathbb{E}(X)=0$, similarly $\mathbb{E}(Y)=0$.

(b) Calculate $\mathbb{E}(XY)$.
Since X is only non-zero if $Y=0$ we have that $XY=0$ with probability 1, therefore $\mathbb{E}(XY)=0$ and consequently $\mathbb{C}\text{ov}(X,Y)=0$.

(c) Are X and Y independent?
If X and Y were independent then $F_{X,Y}(x,y)=F_X(x)F_Y(y)$ for all x,y. Choosing $x=-1, y=-1$ we have that $F_{X,Y}(-1,-1)=0$ however $F_X(-1)=0.25=F_Y(-1)$ so X and Y are not independent.

2. Evaluate the following Integrals:

(a)
$$\int_0^1\int_0^1 x^2+y\,dxdy$$

$$\begin{aligned}\int_0^1\int_0^1 x^2+y\,dxdy &= \int_0^1\left[\frac{1}{3}x^3+yx\right]_{x=0}^{x=1}dy\\ &= \int_0^1 \frac{1}{3}+y\,dy\\ &= \left[\frac{1}{3}y+\frac{1}{2}y^2\right]_{y=0}^{y=1}\\ &= \frac{1}{3}+\frac{1}{2}=\frac{5}{6}.\end{aligned}$$

(b)
$$\int_2^5\int_1^3 xy\,dxdy$$

$$\begin{aligned}\int_2^5\int_1^3 xy\,dxdy &= \int_2^5\left[\frac{1}{2}x^2y\right]_{x=1}^{x=3}dy\\ &= \int_2^5 \frac{1}{2}9y-\frac{1}{2}y\,dy\\ &= \int_2^5 4y\,dy\\ &= \left[2y^2\right]_{y=2}^{y=5}\\ &= 42.\end{aligned}$$

(c)

$$\int_0^1 \int_0^5 (x+y)^2 \, dy \, dx$$

Note the order of integration is y first this time.

$$\begin{aligned}
\int_0^1 \int_0^5 (x+y)^2 \, dy \, dx &= \int_0^1 \left[\frac{1}{3}(x+y)^3\right]_{y=0}^{y=5} dx \\
&= \frac{1}{3}\int_0^1 (5+x)^3 - x^3 dx \\
&= \frac{1}{3}\int_0^1 125 + 75x + 15x^2 dx \\
&= \frac{1}{3}\left[125x + \frac{75}{2}x^2 + 5x^3\right]_{x=0}^{x=1} \\
&= \frac{335}{6}.
\end{aligned}$$

(d)

$$\int_1^2 \int_0^5 \frac{1}{(x+y)^3} \, dydx$$

$$\begin{aligned}
\int_1^2 \int_0^5 \frac{1}{(x+y)^3} \, dydx &= \int_1^2 \left[-\frac{1}{2}(x+y)^{-2}\right]_{y=0}^{y=5} dx \\
&= \int_1^2 \frac{1}{2}x^{-2} - \frac{1}{2}(x+5)^{-2} \, dx \\
&= \left[-\frac{1}{2}x^{-1} + \frac{1}{2}(x+5)^{-1}\right]_{x=1}^{x=2} \\
&= -\frac{1}{4} + \frac{1}{14} + \frac{1}{2} - \frac{1}{12} = \frac{10}{42}.
\end{aligned}$$

(e) Let $R = [0,3] \times [0,5]$

$$\iint_R x \, dxdy$$

$$\begin{aligned}
\iint_R x \, dxdy &= \left(\int_0^5 dy\right)\left(\int_0^3 x \, dx\right) \\
&= 5\left[\frac{1}{2}x^2\right]_{x=0}^{x=3} \\
&= \frac{45}{2}.
\end{aligned}$$

3. Evaluate the following integrals:

(a)

$$\int_0^1 \int_0^x 2x + 5y \, dydx$$

$$\begin{aligned}
\int_0^1 \int_0^x 2x + 5y \, dydx &= \int_0^1 \left[2xy + \frac{5}{2}y^2\right]_{y=0}^{y=x} dx \\
&= \int_0^1 2x^2 + \frac{5}{2}x^2 dx \\
&= \left[\frac{2}{3}x^3 + \frac{5}{6}x^3\right]_{x=0}^{x=1} \\
&= \frac{9}{6} = \frac{3}{2}.
\end{aligned}$$

(b) Let $R = \{(x, y) : 0 \leq x \leq y \leq 2\}$

$$\iint_R e^{x+y}\, dxdy$$

$$\begin{aligned}
\iint_R e^{x+y}\, dxdy &= \int_0^2 \int_0^y e^{x+y}\, dxdy \\
&= \int_0^2 \left[e^{x+y}\right]_{x=0}^{x=y}\, dy \\
&= \int_0^2 e^{2y} - e^y\, dy \\
&= \left[\frac{1}{2}e^{2y} - e^y\right]_{y=0}^{y=2} \\
&= \frac{1}{2}e^4 - e^2 - \left(\frac{1}{2} - 1\right) \\
&= \frac{1}{2}\left(e^4 - 2e^2 + 1\right).
\end{aligned}$$

4. Let $R = \{(x, y) : x^2 + 2y^2 \leq 1, x \geq 0\}$ and evaluate

$$\iint_R x\, dxdy.$$

$$\begin{aligned}
\iint_R x\, dxdy &= \int_0^1 \int_{-\frac{\sqrt{1-x^2}}{\sqrt{2}}}^{\frac{\sqrt{1-x^2}}{\sqrt{2}}} x\, dydx \\
&= \sqrt{2}\int_0^1 x\sqrt{1-x^2}\, dx \\
&= \sqrt{2}\int_0^1 \frac{d}{dx}\left(-\frac{1}{3}(1-x^2)^{\frac{3}{2}}\right) dx \\
&= \frac{\sqrt{2}}{3}\left[-(1-x^2)^{\frac{3}{2}}\right]_{x=0}^{x=1} \\
&= \frac{\sqrt{2}}{3}.
\end{aligned}$$

5. The Random variables X and Y have joint density function

$$f_{X,Y}(x, y) = \frac{1}{y}e^{-y-\frac{x}{y}}, \quad 0 \leq x, y < \infty.$$

(a) Verify that $f_{X,Y}$ is indeed a joint density function by showing that it integrates to 1.

We need to evaluate

$$\int_0^\infty \int_0^\infty \frac{1}{y}e^{-y-\frac{x}{y}}\, dxdy.$$

Observe that the integration with respect to x appears the most straightforward so let us do that first.

$$\int_0^\infty \int_0^\infty \frac{1}{y}e^{-y-\frac{x}{y}}\, dxdy = \int_0^\infty e^{-y} \int_0^\infty \frac{1}{y}e^{-\frac{x}{y}}\, dxdy$$

Observe that for the inner integral y is fixed and $x \mapsto \frac{1}{y}e^{-\frac{x}{y}}$ (for $x \geq 0$ and is zero otherwise) is the probability density function for an exponential random variable with rate $1/y$, therefore it integrates to 1. This gives

$$\int_0^\infty \int_0^\infty \frac{1}{y} e^{-y-\frac{x}{y}}\, dxdy = \int_0^\infty e^{-y}\, dy$$

What remains is the integral of a probability density function for an exponential random variable with rate 1, so this also integrates to 1. Therefore we have

$$\iint_{\mathbb{R}^2} f_{X,Y}(x,y)dxdy = 1$$

and hence $f_{X,Y}$ is a valid joint probability density function.
Note that here (X, Y) could be obtained by letting $Y \sim \text{Exp}(1)$ then we let $X \sim \text{Exp}(1/Y)$.

(b) As in the previous part we will choose y for the outer integral so

$$\begin{aligned}
\mathbb{P}(X \leq Y) &= \int_0^\infty \int_0^y \frac{1}{y} e^{-y-\frac{x}{y}}\, dx\, dy \\
&= \int_0^\infty e^{-y} \int_0^y \frac{d}{dx}\left(-e^{-\frac{x}{y}}\right) dx\, dy \\
&= \int_0^\infty e^{-y}\left(1 - e^{-1}\right) dy
\end{aligned}$$

To obtain the final line we have used the fundamental theorem of calculus. What remains is

$$\mathbb{P}(X \leq Y) = \left(1 - e^{-1}\right) \int_0^\infty e^{-y} dy.$$

Observe that $\int_0^\infty e^{-y}dy$ is the integral of the probability density function for a Exp(1)-random variable over its image and hence is equal to 1, giving

$$\mathbb{P}(X \leq Y) = \left(1 - e^{-1}\right).$$

6. We are given

$$f_{X,Y} = \begin{cases} \frac{2}{b}e^{-2x}, & x > 0 \text{ and } 0 \leq y \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

So the probability $\mathbb{P}(Y - X > 1)$ involves integrating the probability density function over the region $A = \{(x, y) : y - x > 1, x > 0, 0 \leq y \leq b\}$ which is the intersection of the set $\{(x, y) : y - x > 1\}$ with the set of all points (x, y) for which $f_{X,Y}$ is non-zero. This area is given in Figure 1 below. We will use y for the outer integral which (we see from the plot) varies between 1 and b. For each fixed value of y, x varies from 0 to $y - 1$, that is

$$\begin{aligned}
\mathbb{P}(Y - X > 1) &= \int_{y=1}^{b} \int_{x=0}^{y-1} \frac{2}{b} e^{-2x}\, dx\, dy \\
&= \frac{1}{b} \int_{y=1}^{b} \left[-e^{-2x}\right]_{x=0}^{y-1} dy \\
&= \frac{1}{b} \int_{y=1}^{b} 1 - e^{-2(y-1)}\, dy \\
&= \frac{1}{b} \left[y + \frac{1}{2} e^{-2(y-1)}\right]_{y=1}^{b} \\
&= \frac{1}{b} \left(b - 1 + \frac{1}{2}(e^{-2(b-1)} - 1)\right) \\
&= 1 - \frac{3}{2b} + \frac{1}{2b} e^{-2(b-1)}.
\end{aligned}$$

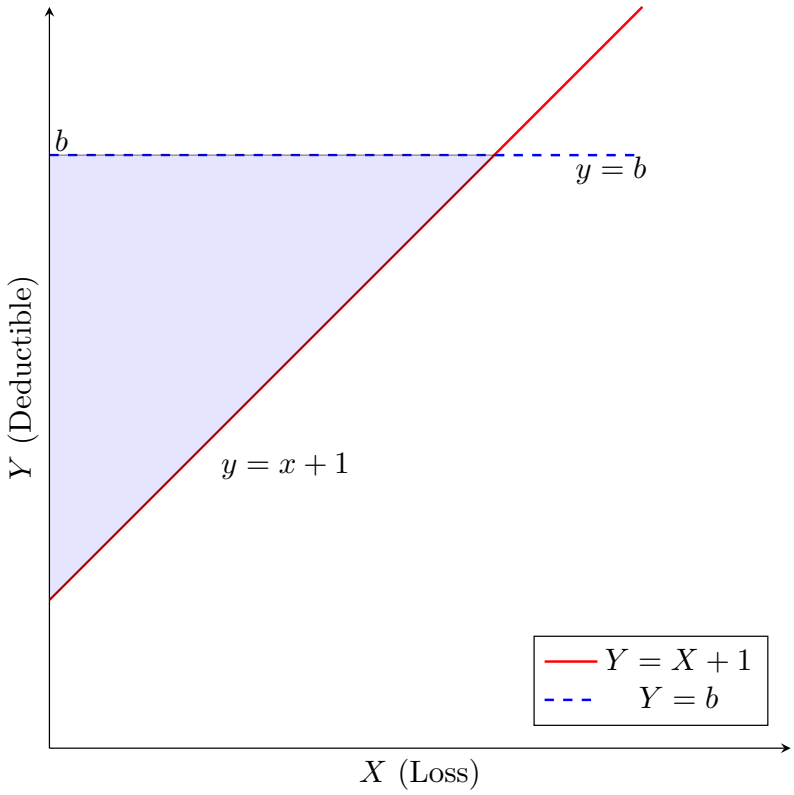


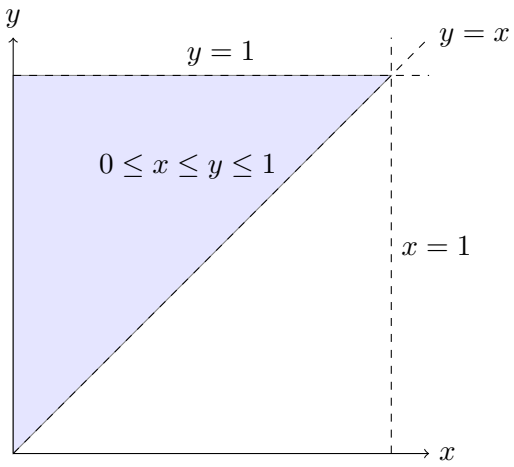
Figure 1: Area of integration for Q6

Exam Style Question

(a) To verify that $f(x, y) = 6(1 - y)$ is a valid joint probability density function, we check:

(i) Non-negativity: For $0 \leq y \leq 1$, we have $1 - y \geq 0$, so $f(x, y) = 6(1 - y) \geq 0$. Thus, the function is non-negative over its support.

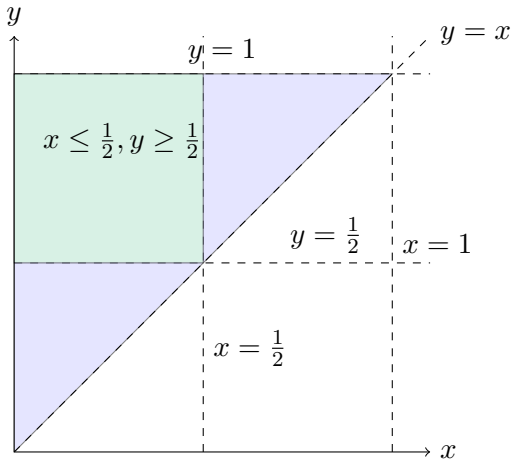
(ii) Normalization: We start by plotting the area of integration:



$$\begin{aligned}\int_{y=0}^{1}\int_{x=0}^{y} 6(1-y)\,dx\,dy &= \int_{y=0}^{1} 6(1-y)\cdot y\,dy \\ &= 6\int_{y=0}^{1} y(1-y)\,dy \\ &= 6\left[\frac{y^2}{2}-\frac{y^3}{3}\right]_{y=0}^{1} \\ &= 6\left(\frac{1}{2}-\frac{1}{3}\right)=1\end{aligned}$$

So $f(x,y)$ is a valid joint probability density function. [3 marks]

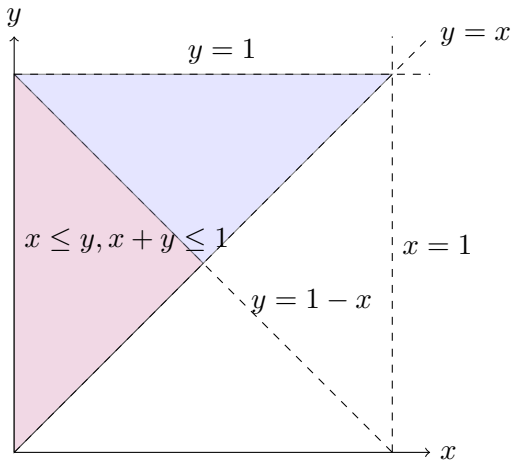
(b)



$$\begin{aligned}\int_{y=\frac{1}{2}}^{1}\int_{x=0}^{\frac{1}{2}} 6(1-y)\,dx\,dy &= \int_{y=\frac{1}{2}}^{1} 6(1-y)\cdot\frac{1}{2}\,dy \\ &= 3\int_{y=\frac{1}{2}}^{1} (1-y)\,dy \\ &= 3\left[y-\frac{y^2}{2}\right]_{y=\frac{1}{2}}^{1} \\ &= 3\left(\frac{1}{2}-\frac{3}{8}\right)=\frac{3}{8}\end{aligned}$$

[4 marks]

(c)



$$\begin{aligned}\int_{x=0}^{\frac{1}{2}}\int_{y=x}^{1-x} 6(1-y)\,dy\,dx &= \int_{x=0}^{\frac{1}{2}} 6\left[y-\frac{y^2}{2}\right]_x^{1-x} dx \\ &= \int_{x=0}^{\frac{1}{2}} 6\left(\frac{1}{2}-x\right) dx \\ &= 6\left[\frac{x}{2}-\frac{x^2}{2}\right]_{x=0}^{\frac{1}{2}} \\ &= 6\left(\frac{1}{4}-\frac{1}{8}\right) = \frac{3}{4}\end{aligned}$$

[3 marks]